

LVM , postgres and Zabbix server installation and my challengs.

In this sampel we are going to explore some exciting commands! First we must add a new storage to our virtual machine. I'm using UTM and ubuntu 24.04 so I added the new space with UTM.

After that, they asked us (respecdetly :D) to create a Logical voloum using LVM. Next we are going to install postgres and run a Zabbix server. Here I h've had installed Ngingx as Web Server already.

First setep , Fdisk and LVM

Exploring present storage system using lsblk

```
lsblk
```

This command show you your current storage architecure :

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
vda	253:0	0	40G	0	disk	
vda1	253:1	0	1G	0	part	/boot
vda2	253:2	0	2G	0	part	
ubuntu-vg-root						
	252:0	0	37G	0	lvm	/
vdb	253:16	0	20G	0	disk	

Reformatting the partion using fdisk

```
sudo fdisk /dev/vdb
```

```
n
```

```
p
```

```
1
```

```
<Enter>
```

```
<Enter>
```

```
t
```

```
8e
```

```
w
```

- n → new partition

- p → primary partition
- 1 → partition number
- Enter → first available sector
- Enter → last available sector (use whole disk)
- t → change partition type
- 8e → Linux LVM (on MBR partition tables)
- w → write changes and exit

Now we are going to check the new partition

Verify the partition

```
lsblk
```

Output:

```
vdb
  vdb1
```

So everything is fine! let's

Let's use LVM and create our physical volume first

```
sudo pvcreate /dev/vdb1
```

Output:

Physical volume "/dev/vdb1" successfully created.

Create a Volume Group

```
sudo vgcreate data-vg /dev/vdb1
```

Output:

Volume group "data-vg" successfully created

Verify:

```
sudo vgs
```

Output:

VG	#PV	#LV	#SN	Attr	VSize	VFree
data-vg	1	0	0	wz--n-	20.00g	20.00g

Create a Logical Volume

Use the entire VG:

```
sudo lvcreate -l 100%FREE -n data-lv data-vg
```

Output:

Logical volume "data-lv" created.

Verify:

```
sudo lvs
```

Output:

LV	VG	Attr	LSize
data-lv	data-vg	-wi-a-----	20.00g

Last command to make it mountable

```
sudo mkfs.ext4 /dev/data-vg/data-lv
```

So now we have a volume which is mountable for our data directory. Next, I installed postgresql and after that I assign the generated partition for its data.

```
sudo apt install postgresql postgresql-contrib
```

After installing postgresql, first I stop the database with systemctl and made it ready for migration.

```
systemctl status postgresql  
systemctl disable --now postgresql
```

The next code will show you postgres data directory which was used when it was active.

```
sudo -u postgres psql -c "SHOW data_directory;"
```

Now we are going to mount our new partition and change postgres data directory.

So we use rsync to copy everything inside postgres previous data directory and also keep a backup.

```
mkdir /mnt/tem_new_pg
mount /dev/data-vg/data-lv /mnt/tem_new_pg/
rsync -aHAX /var/lib/postgresql/ /mnt/tem_new_pg/
mv /var/lib/postgresql /var/lib/postgresql.bk
mkdir /var/lib/postgresql
umount /mnt/tem_new_pg
mount /dev/data-vg/data-lv /var/lib/postgresql
```

Now we are going to make our postgres ready

```
chown -R postgres:postgres /var/lib/postgresql
systemctl status postgresql
systemctl enable postgresql
```

To check everything is fine, we can use df to check our disk health. Also with blkid we are going to get the UUID of our storage. We are going to add its UUID to fstab which will mount automatically our storage space in next reboot.

Installing Zabbix server

In next part I installed the zabbix server for nginx. I had to first get the deb with wget

```
wget https://repo.zabbix.com/zabbix/7.0/ubuntu-arm64/pool/main/z/zabbix-release/zabbix-release_7.0-1_all.deb
dpkg -i zabbix-release_latest_7.0+ubuntu24.04_all.deb
```

Tips : make sure to download the compatible version with your ubuntu and your CPU architecture.

After building the installer, we are going to install Zabbix

I'm going to install the server compatible with my nginx.

```
sudo dpkg -i zabbix-release_latest_7.0+ubuntu24.04_all.deb
sudo apt install zabbix-server-pgsql
zabbix-frontend-php
zabbix-nginx-conf
```

```
zabbix-sql-scripts
zabbix-agent2
```

```
zcat /usr/share/zabbix-sql-scripts/postgresql/server.sql.gz | psql -U zabbix -d zabbix
/etc/zabbix/zabbix_server.conf
```

Set

```
DBHost=localhost
DBName=zabbix
DBUser=zabbix
DBPassword=StrongPasswordHere
```

After that , we are going to add our Zabbix and connect to our postgres

eSQL:

```
CREATE USER zabbix WITH PASSWORD 'YourStrongPassword';
CREATE DATABASE zabbix OWNER zabbix;
\q
```

Fist challnge :

I wanted to see Zabbix dashboard on my localhost/Zabbix address. for make it work with nginx , first I disabled the Zabbix.conf and added a new location inside mu nginx config

The code is :

```
sudo mv /etc/nginx/conf.d/zabbix.conf /etc/nginx/conf.d/zabbix.conf.disabled
```

Adding new config with vim in nginx.conf

```
location /zabbix {
    alias /usr/share/zabbix;
    index index.php;

    location ~ ^/zabbix/(.+\.php)$ {
        alias /usr/share/zabbix/$1;

        fastcgi_pass unix:/var/run/php/zabbix.sock;
        fastcgi_index index.php;
    }
}
```

```

        include fastcgi_params;

        fastcgi_param SCRIPT_FILENAME /usr/share/zabbix/$1;
        fastcgi_param DOCUMENT_ROOT /usr/share/zabbix;
    }

    location ~ ^/zabbix/(.+\. (css|js|png|jpg|gif|svg|ico))$ {
        alias /usr/share/zabbix/$1;
    }

    try_files $uri $uri/ /zabbix/index.php;
}

```

nginx -t to test new config and after that , enable all services using systemctl and restart them

Boom! now we can see our dashboard installation in /localhost/Zabbix

Second challenge in installing dashboard :

In my installation I had issue with connecting to my database. After some reasearch and asking from Gpt , the final values for host and port was this :

Database host: 127.0.0.1 Port: 5432

Question : Why using LVM ?

With LVM

LVM adds a layer of abstraction.

Physical Disk	Physical Volume (PV)	Volume Group (VG)
Logical Volume	Logical Volume	Filesystem Filesystem

Instead of partitions owning fixed sizes, all storage goes into a pool (the Volume Group), and Logical Volumes are created from that pool.